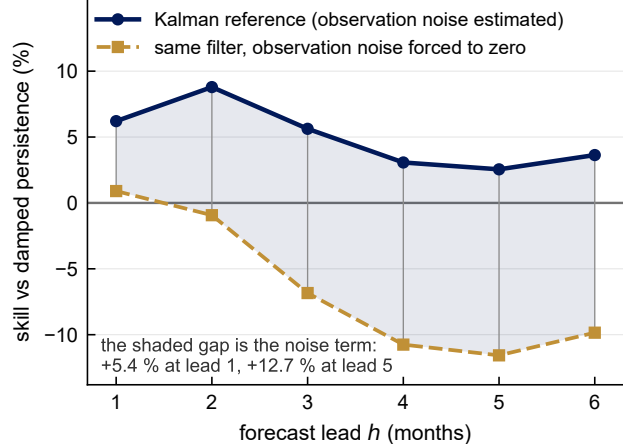
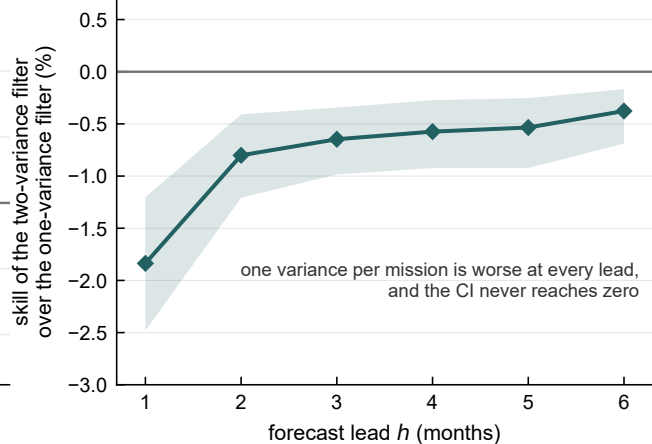


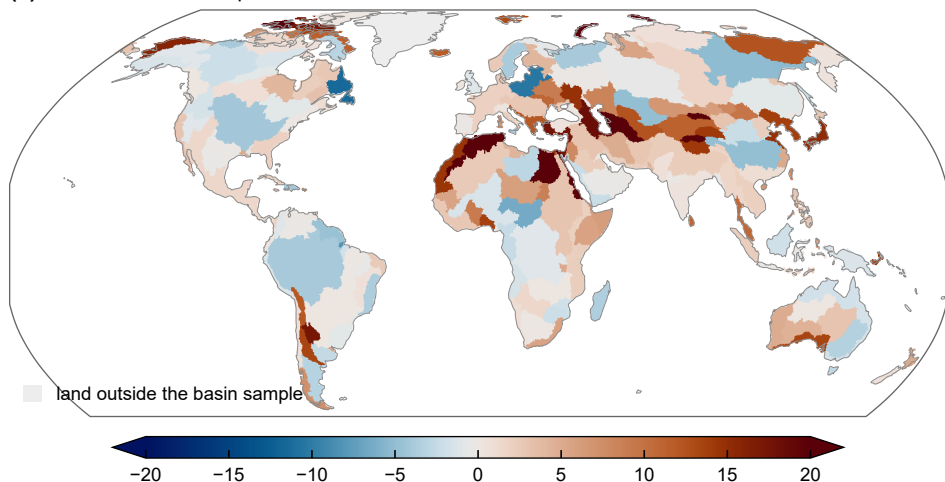
(a) the observation-noise term



(b) a separate GRACE-FO noise variance



(c) where the filter helps: better in 158 of 234 basins, worse in 76



per-basin lead-1 skill of the Kalman reference over damped persistence, AR(1) variant (%), from RMSE in cm
red: the filter is better; blue: damped persistence is better; scale clipped at ± 20 %